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OCI-Dedicated KMS Activity Log user experience

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Safe Harbor Statement

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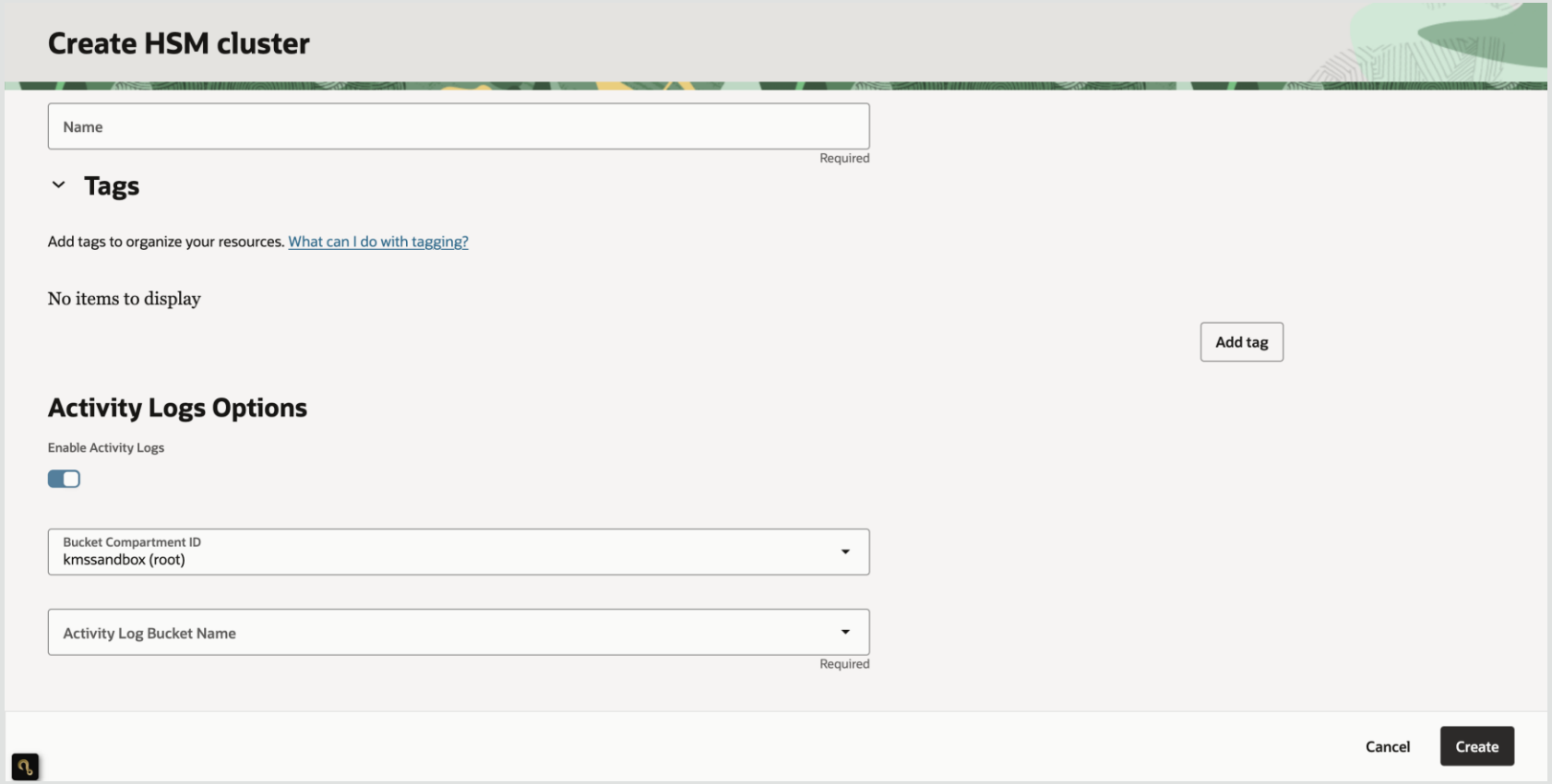
Scope:

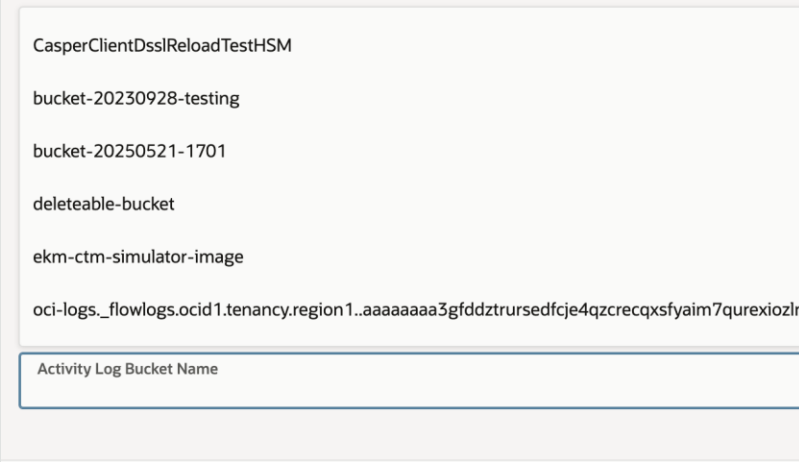
The Activity log will consist of

- **HSM Configuration and Initialization Logs:** Logs that detail events related to the configuration and initialization of the HSM.
- **Key Lifecycle Management (LCM) Operations Logs in the HSM:** Detailed records that show the LCM operations in the HSM.
- **User activity:** Logs necessary to update actions performed in the HSM. (Aakash Chaudhary confirms this Includes Users login to HSM and Actions Performed by the User on HSM).

During DKMS cluster creation:

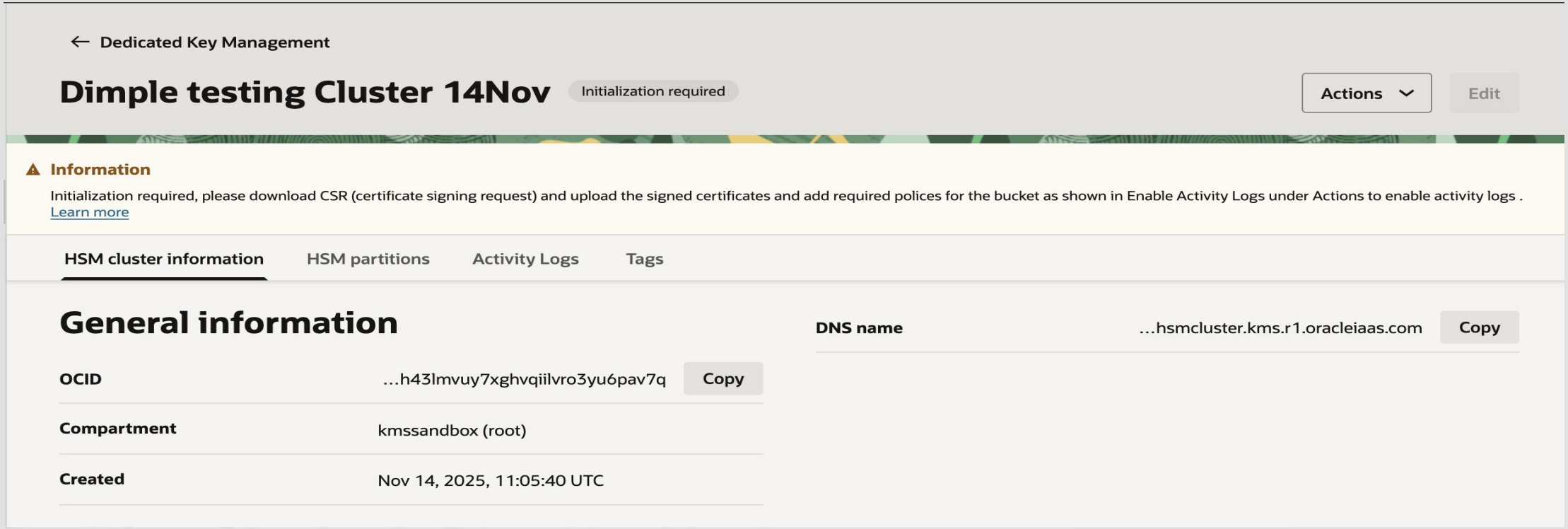
- OCI will ask for Compartment and Bucket information during cluster creation workflow





Cluster goes in intialization required status

- Along with download CSR and upload signed certificates, customer is also asked to add the polices for activity logs when the cluster is in initialisation required state
- Untill the polices are added the cluster stays in Initialisation Required state.



Customer is asked to check "Enable activity log" action

Information

Initialization required, please download CSR (certificate signing request) and upload the signed certificates and add required polices for the bucket as shown in Enable Activity Logs under Actions to enable activity logs .
Learn more

Actions

Edit

able Activity Logs unde

..hsmcluster.kms.r1

- Move resource
- Enable Activity Logs
- Add tags
- Delete HSM cluster
- Initialize cluster

Customer is asked to create the polices and click enable activity logs

Enable Activity Logs

Current State	WAITING_FOR_CUSTOMER
Activity Log Bucket Name	audit_logs_validation
Bucket Namespace	kmstest
Compartment ID	ocid1.compartment.region1.sea.amaaaaaa62pi5waat6pqguz7by3yhgp7w4di3uvt7o72lko2c5cq6efocd3q

Please Complete all the pending instructions as shown and click on ENABLE ACTIVITY LOGS once the audit logging state moves to WAITING_FOR_CUSTOMER

1. Create a dynamic group named **DKMSAuditLogUploader-avuro6caaae4o**
with the matching rule:
ALL {resource.type='instance', resource.tag.namespace='oracle.dkms', resource.tag.key='cluster-ocid', resource.tag.value='ocid1.kmshsmcluster.region1.sea.avuro6caaae4o.abzwkljs26vutpnbewgiqguv64m2wqfq3al22rpe6qs6nuitwelkxucpnavq'}.
2. Add a bucket policy to allow the dynamic group to manage objects in the bucket **audit_logs_validation**
with the policy statement as :
Allow dynamic-group DKMSAuditLogUploader-avuro6caaae4o to manage objects in tenancy where all { target.bucket.name='audit_logs_validation', any { request.permission='OBJECT_CREATE', request.permission='OBJECT_INSPECT' } }.

Cancel | Enable Activity Logs

Activity log status changes to verifying post policy creation

Enable Activity Logs

Current State	Verifying
Activity Log Bucket Name	audit_logs_validation
Bucket Namespace	kmstest
Compartment ID	ocid1.compartment.region1.sea.amaaaaaa62pi...

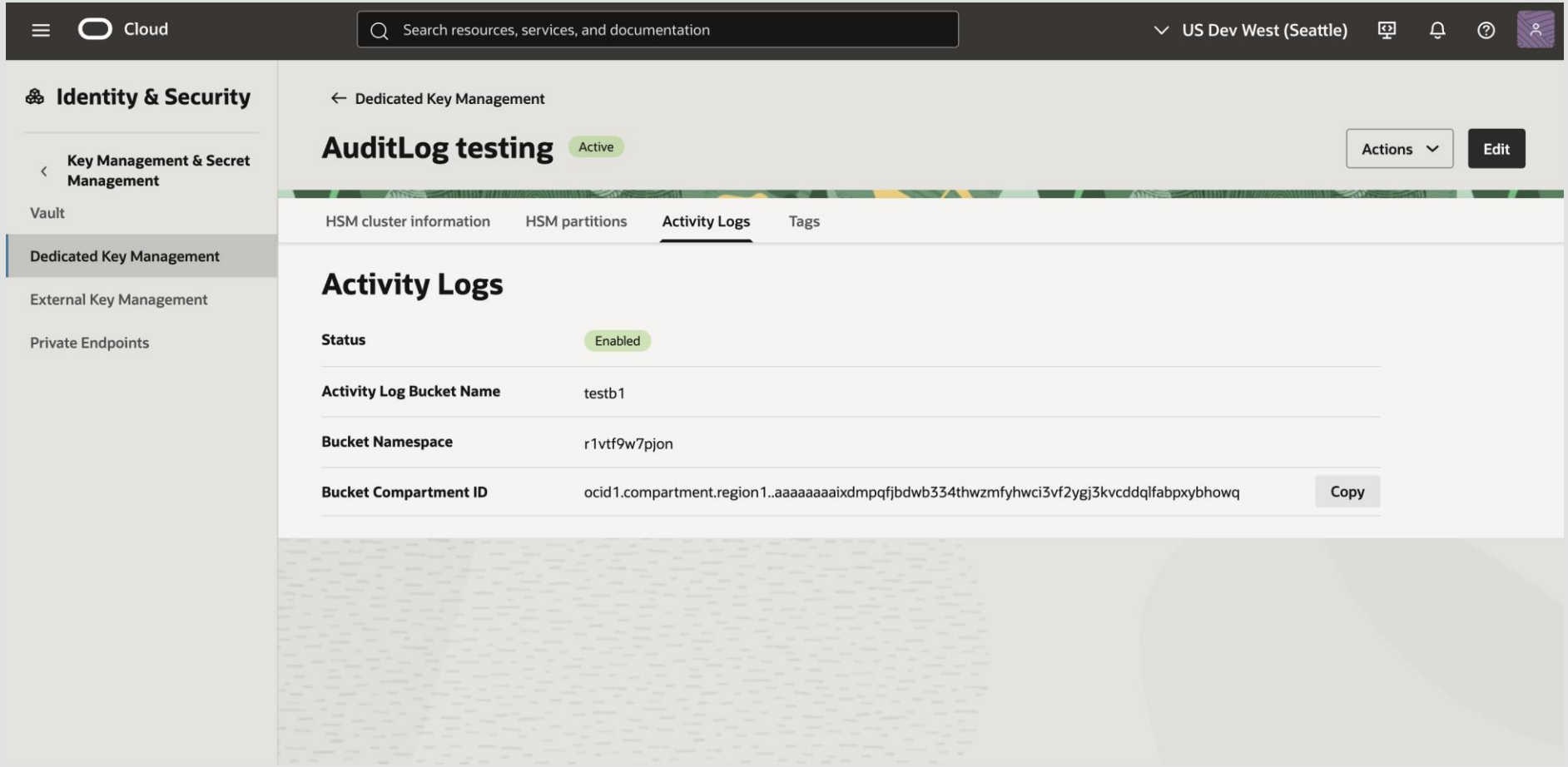
Please Complete all the pending instructions as shown and click on ENABLE ACTIVITY LOGS once the audit logging state moves to WAITING_FOR_CUSTOMER

1. Create a dynamic group named **DKMSAuditLogUploader-avuro6caaae4o** with the matching rule:
ALL {resource.type='instance', resource.tag.namespace='oracle.dkms', resource.tag.key='cluster-ocid', resource.tag.value='ocid1.kmshsmcluster.region1.sea.avuro6caaae4o.abzwkljs26vutpnbewgiqguv64m2wqfq3al22rpe6qs6nuitwelkxucpnavq'}.
2. Add a bucket policy to allow the dynamic group to manage objects in the bucket **audit_logs_validation**
with the policy statement as :
Allow dynamic-group DKMSAuditLogUploader-avuro6caaae4o to manage objects in tenancy

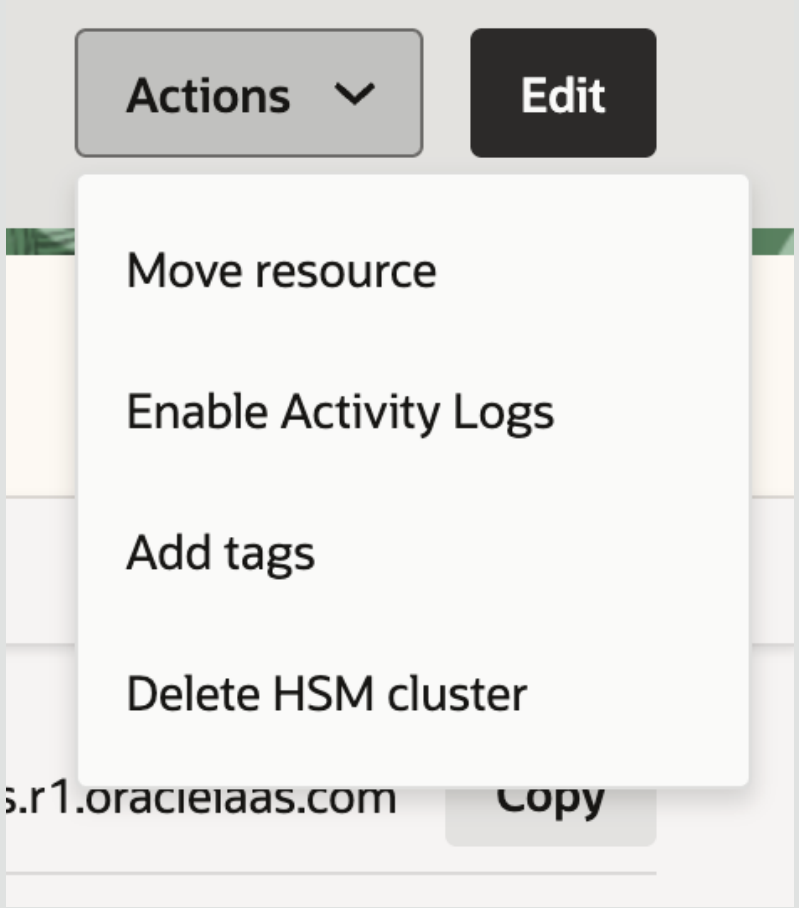
Cancel

Enable Activity Logs

Cluster is in active state and activity logs enabled.

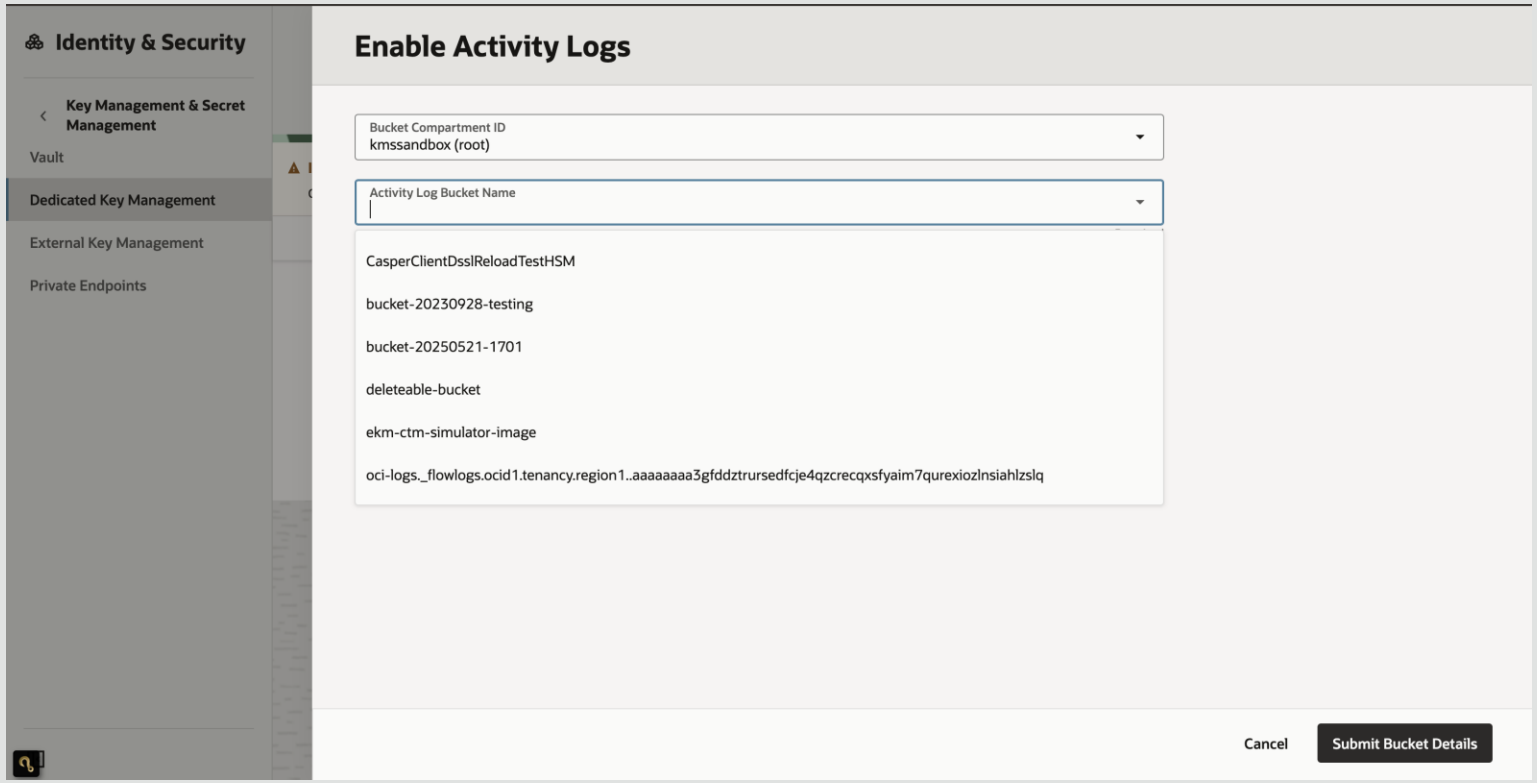


Enabling activity log for exiting DKMS clusters



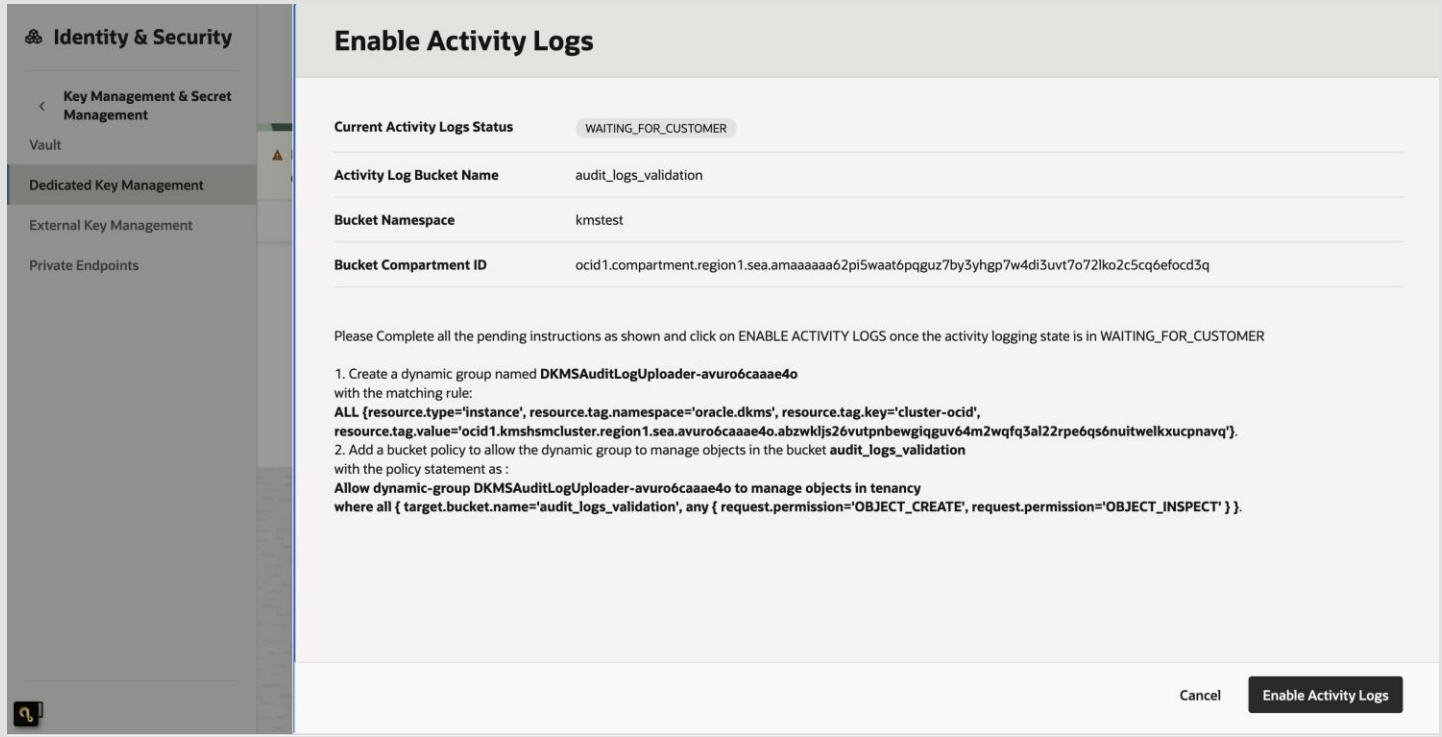
Enabling activity log for exiting DKMS clusters:

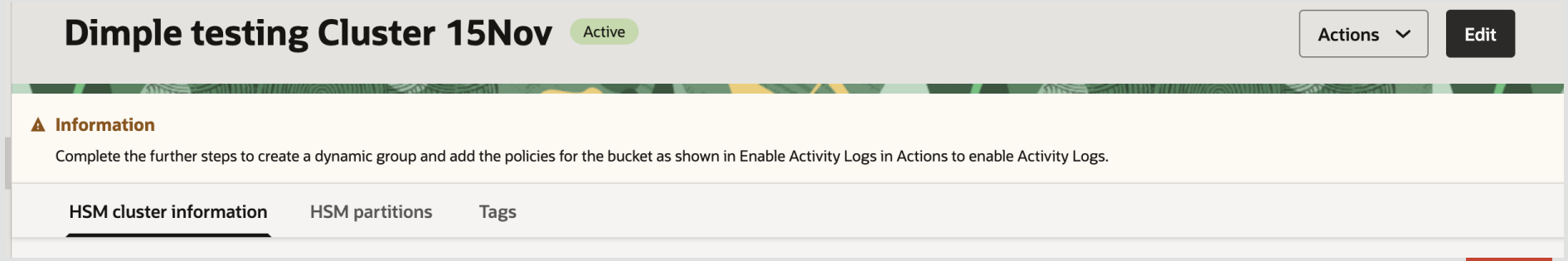
The customer will be asked for compartment and bucket information



Enabling activity log for exiting DKMS clusters: cntd…

Based on the selection the customer will be asked to complete policy creation.





Activity log status changes to verifying post policy creation

Enable Activity Logs

Current State	Verifying
Activity Log Bucket Name	audit_logs_validation
Bucket Namespace	kmstest
Compartment ID	ocid1.compartment.region1.sea.amaaaaaa62pi...

Please Complete all the pending instructions as shown and click on ENABLE ACTIVITY LOGS once the audit logging state moves to WAITING_FOR_CUSTOMER

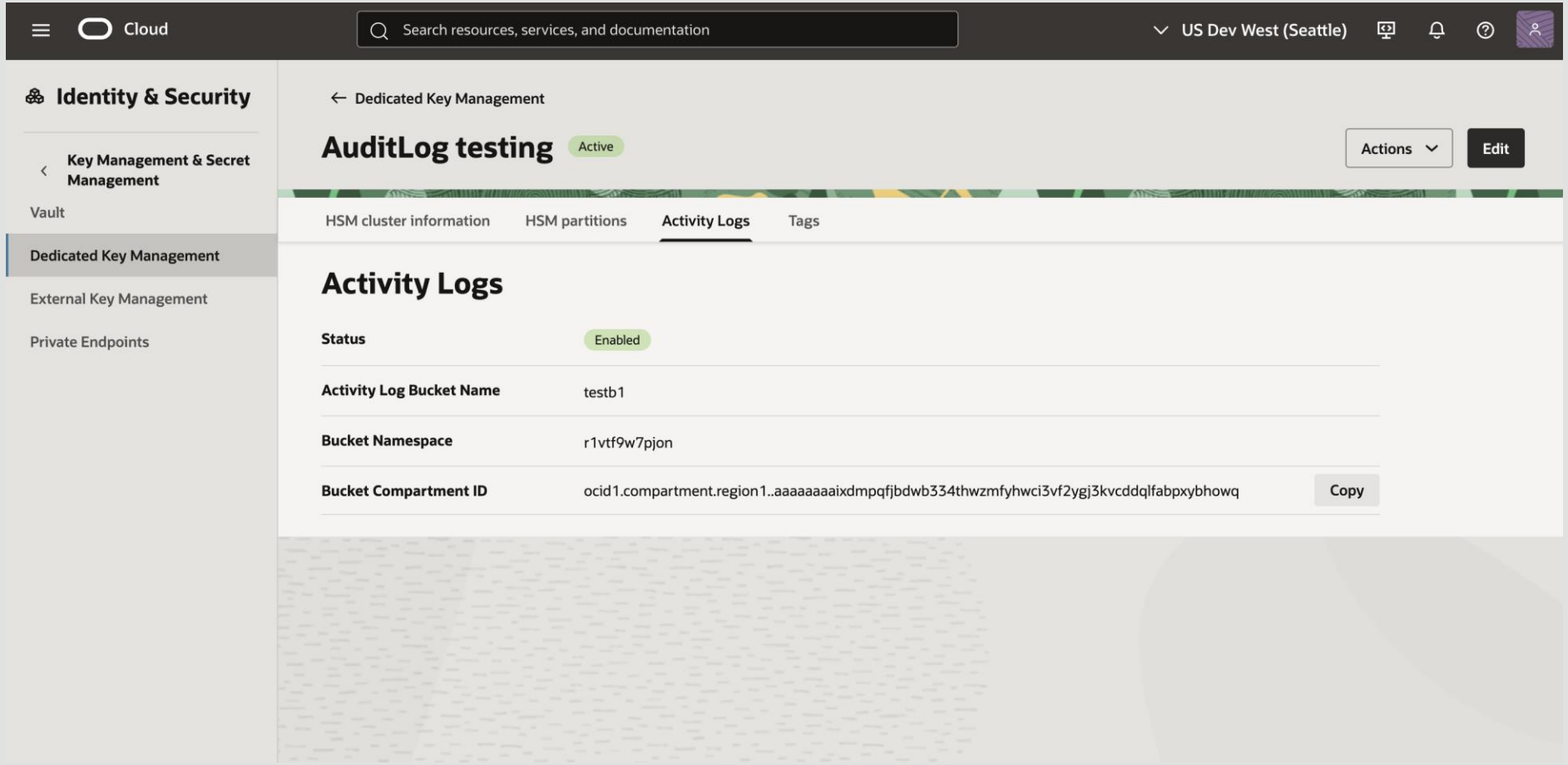
1. Create a dynamic group named **DKMSAuditLogUploader-avuro6caaae4o** with the matching rule:
ALL {resource.type='instance', resource.tag.namespace='oracle.dkms', resource.tag.key='cluster-ocid', resource.tag.value='ocid1.kmshsmcluster.region1.sea.avuro6caaae4o.abzwkljs26vutpnbewgiqguv64m2wqfq3al22rpe6qs6nuitwelkxucpnavq'}.
2. Add a bucket policy to allow the dynamic group to manage objects in the bucket **audit_logs_validation**
with the policy statement as :
Allow dynamic-group DKMSAuditLogUploader-avuro6caaae4o to manage objects in tenancy

Cancel

Enable Activity Logs

Enabling activity log for exiting clusters: cntd…

Once completed customer can continue to enable activity logs for DKMS cluster.



Listing the activity logs from bucket

← Buckets

audit_logs_validation

Bucket

Actions | Upload objects

Details | Objects | Management | Monitoring | Policies | Work requests | Tags

Objects

Search object by name or prefix | Search

Upload objects | Actions

Name/Prefix	Last modified	Size	Storage tier
P10_1763623536.bin	Nov 24, 2025, 08:28 UTC	26.25 KiB	Standard
P10_1763623536.sign	Nov 24, 2025, 08:28 UTC	264 bytes	Standard
P10_1763623587.bin	Nov 24, 2025, 08:28 UTC	256 bytes	Standard
P10_1763623587.sign	Nov 24, 2025, 08:28 UTC	264 bytes	Standard
test.sign	Nov 20, 2025, 04:46 UTC	54 bytes	Standard

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Viewing the activity logs

How Customers Use Cfm2AuditLogUtil

- Download Required Files : download the audit log (`.bin`), signature (`.sign`), and certificate files from Object Storage bucket.
- Use the CLI to convert and verify logs, using Cfm2AuditLogUtil
- **Command : Cfm2AuditLogUtil \ -b P0_1721037000.bin # required -t P0_1721037000.txt # required -s P0_1721037000.sign # optional -pcert po_cert.pem # optional, with -s -hcert hsm_cert.pem # optional, with –s**

  .bin (Audit Log File): Contains the raw, immutable binary audit log entries generated by the Marvell HSM firmware.
- **.sign (Signature File): Contains the cryptographic RSA-2048 signature over the** `.bin` file, *Used to prove the authenticity and completeness of the audit log file.*
- After a successful run, the specified output text file (**`.txt`**) will contain a human-readable, pretty-printed version of the logs.

Thank you